

CSCI3150 Introduction to Operating Systems

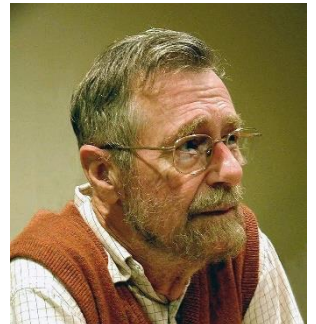
Lecture 7: Synchronization 3: Semaphores

Hong Xu

<https://github.com/henryhxu/CSCI3150>

Semaphore

- ▣ It provides
 - ◆ Mutex
 - ◆ and atomic counters
- ▣ Two operations:
 - ◆ `P(semaphore), wait()`: after the Dutch word for test
 - ◆ `V(semaphore), signal(), post()`: after the Dutch word for increment
- ▣ Probably the most unintuitive names you encounter in this course
 - ◆ You have *Edsger W. Dijkstra* to thank to...



Semaphore: A Definition

- An object with an integer value and two routines

- ◆ `sem_wait()` and `sem_post()`.

- ◆ Initialization

```
1  #include <semaphore.h>
2  sem_t s;
3  sem_init(&s, 0, 1); // initialize s to the value 1
```

- Declare a semaphore `s` and initialize it to the value 1
 - The second argument, 0, indicates that the semaphore is shared between *threads in the same process*.

Interact with Semaphore

▣ `sem_post()`

```
1  int sem_post(sem_t *s) {  
2      increment the value of semaphore s by one  
3      if there are one or more threads waiting, wake one  
4  }
```

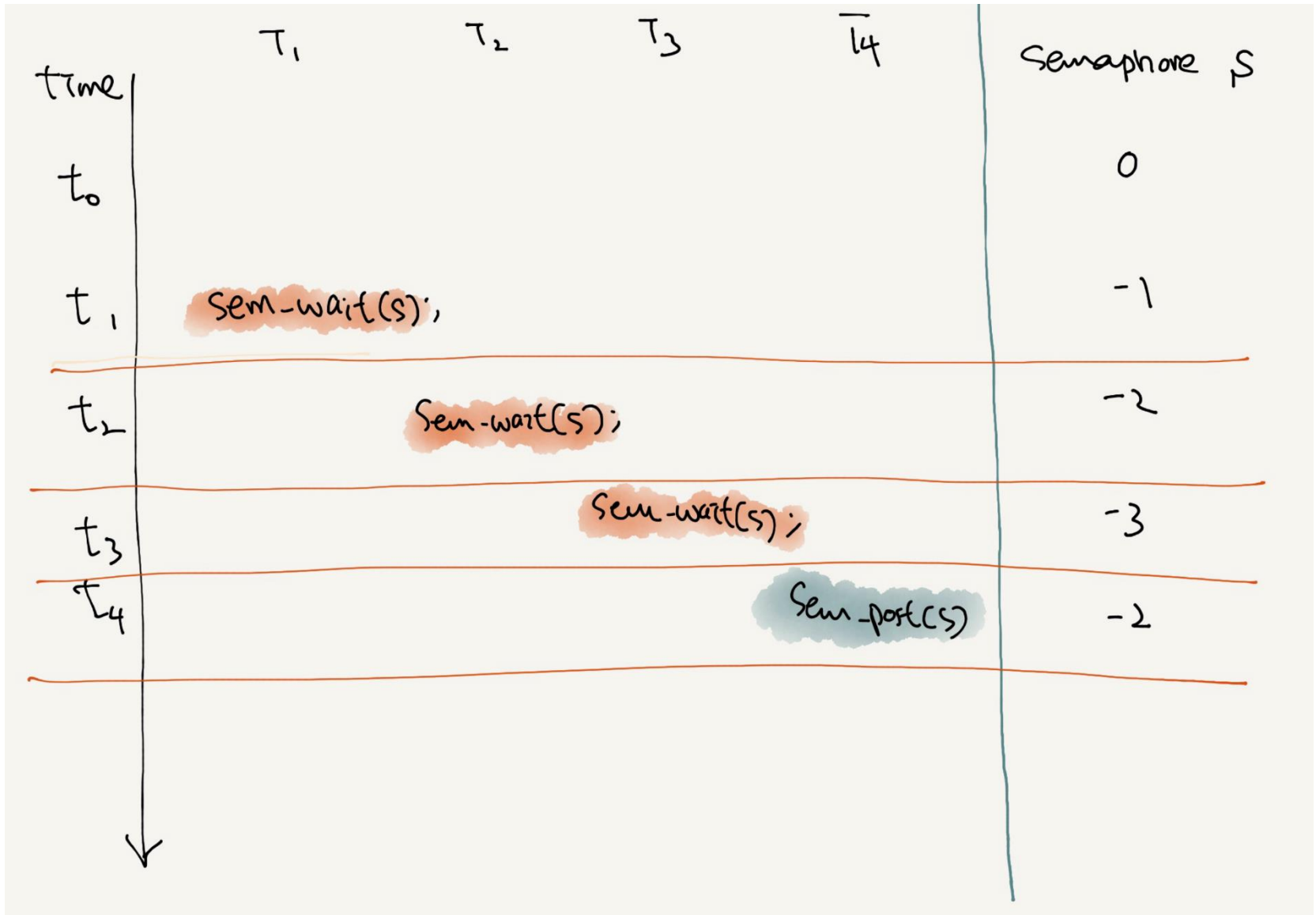
- ◆ Simply **increments** the value of the semaphore.
- ◆ If there is **at least** a thread waiting, **wakes one** of them up.
 - This happens when the current value is negative or zero

Interact with Semaphore

▣ `sem_wait()`

```
1  int sem_wait(sem_t *s) {  
2      decrement the value of semaphore s by one  
3      wait if value of semaphore s is negative  
4  }
```

- ◆ If the value of the semaphore was *one* or *higher* when called `sem_wait()`, **return right away**.
- ◆ Otherwise it will cause the caller to suspend execution waiting for a subsequent post.
- ◆ When negative, the value of the semaphore is equal to the number of waiting threads.



Binary Semaphores (Locks)

▣ What should **x** be?

- ◆ The initial value should be **1**.

```
1  sem_t m;  
2  sem_init(&m, 0, X); // initialize semaphore to X; what should X be?  
3  
4  sem_wait(&m);  
5  //critical section here  
6  sem_post(&m);
```

Value of Semaphore	Thread 0	Thread 1
1		
1	call sem_wait()	
0	sem_wait() returns	
0	(crit sect)	
0	call sem_post()	
1	sem_post() returns	

Thread Trace: Two Threads Using A Semaphore

Value	Thread 0	State	Thread 1	State
1		Running		Ready
1	call sem_wait()	Running		Ready
0	sem_wait() returns	Running		Ready
0	(crit set: begin)	Running		Ready
0	<i>Interrupt; Switch → T1</i>	Ready		Running
0		Ready	call sem_wait()	Running
-1		Ready	decrement sem	Running
-1		Ready	(sem < 0) → sleep	sleeping
-1		Running	<i>Switch → T0</i>	sleeping
-1	(crit sect: end)	Running		sleeping
-1	call sem_post()	Running		sleeping
0	increment sem	Running		sleeping
0	wake(T1)	Running		Ready
0	sem_post() returns	Running		Ready
0	<i>Interrupt; Switch → T1</i>	Ready		Running
0		Ready	sem_wait() returns	Running
0		Ready	(crit sect)	Running
0		Ready	call sem_post()	Running
1		Ready	sem_post() returns	Running

Binary Semaphores (Locks)

- ▣ What happens in the previous example with more than 2 threads?

Semaphores As Condition Variables

```
1  sem_t s;
2
3  void *
4  child(void *arg) {
5      printf("child\n");
6      sem_post(&s); // signal here: child is done
7      return NULL;
8  }
9
10 int
11 main(int argc, char *argv[]) {
12     sem_init(&s, 0, X); // what should X be?
13     printf("parent: begin\n");
14     pthread_t c;
15     pthread_create(c, NULL, child, NULL);
16     sem_wait(&s); // wait here for child
17     printf("parent: end\n");
18     return 0;
19 }
```

A Parent Waiting For Its Child

```
parent: begin
child
parent: end
```

The execution result

- ◆ What should **x** be?
 - The value of semaphore should be set to is **0**.

Thread Trace: Parent Waiting For Child (Case 1)

- The parent calls `sem_wait()` before the child has called `sem_post()`.

Value	Parent	State	Child	State
0	Create(Child)	Running	<i>(Child exists; is runnable)</i>	Ready
0	call <code>sem_wait()</code>	Running		Ready
-1	decrement sem	Running		Ready
-1	$(sem < 0) \rightarrow \text{sleep}$	sleeping		Ready
-1	<i>Switch</i> →Child	sleeping	child runs	Running
-1		sleeping	call <code>sem_post()</code>	Running
0		sleeping	increment sem	Running
0		Ready	wake(Parent)	Running
0		Ready	<code>sem_post()</code> returns	Running
0		Ready	<i>Interrupt; Switch</i> →Parent	Ready
0	<code>sem_wait()</code> returns	Running		Ready

Thread Trace: Parent Waiting For Child (Case 2)

- The child runs to completion before the parent call `sem_wait()`.

Value	Parent	State	Child	State
0	Create (Child)	Running	<i>(Child exists; is runnable)</i>	Ready
0	<i>Interrupt; switch→Child</i>	Ready	child runs	Running
0		Ready	call <code>sem_post()</code>	Running
1		Ready	increment sem	Running
1		Ready	wake (nobody)	Running
1		Ready	<code>sem_post()</code> returns	Running
1	parent runs	Running	<i>Interrupt; Switch→Parent</i>	Ready
1	call <code>sem_wait()</code>	Running		Ready
0	decrement sem	Running		Ready
0	$(sem < 0) \rightarrow \text{awake}$	Running		Ready
0	<code>sem_wait()</code> returns	Running		Ready

The Producer/Consumer (Bounded-Buffer) Problem

❑ **Producer:** `put()` interface

- ◆ Wait for a buffer to become *empty* in order to put data into it.

❑ **Consumer:** `get()` interface

- ◆ Wait for a buffer to become *filled* before using it.

```
1  int buffer[MAX];
2  int fill = 0;
3  int use = 0;
4
5  void put(int value) {
6      buffer[fill] = value;    // line f1
7      fill = (fill + 1) % MAX; // line f2
8  }
9
10 int get() {
11     int tmp = buffer[use];    // line g1
12     use = (use + 1) % MAX;    // line g2
13     return tmp;
14 }
```

The Producer/Consumer (Bounded-Buffer) Problem

```
1  sem_t empty;
2  sem_t full;
3
4  void *producer(void *arg) {
5      int i;
6      for (i = 0; i < loops; i++) {
7          sem_wait(&empty);           // line P1
8          put(i);                     // line P2
9          sem_post(&full);            // line P3
10     }
11 }
12
13 void *consumer(void *arg) {
14     int i, tmp = 0;
15     while (tmp != -1) {
16         sem_wait(&full);             // line C1
17         tmp = get();                 // line C2
18         sem_post(&empty);            // line C3
19         printf("%d\n", tmp);
20     }
21 }
22 ...
```

First Attempt: Adding the Full and Empty Conditions

The Producer/Consumer (Bounded-Buffer) Problem

```
21  int main(int argc, char *argv[]) {
22      // ...
23      sem_init(&empty, 0, MAX);           // MAX buffers are empty to begin with...
24      sem_init(&full, 0, 0);              // ... and 0 buffer is full
25      // ...
26  }
```

First Attempt: Adding the Full and Empty Conditions (Cont.)

- ◆ Imagine that MAX is greater than 1
 - If there are multiple producers, **race condition** can happen.
 - It means that the old data there is overwritten.
- ◆ We've forgotten here is **mutual exclusion**.
 - The filling of a buffer and incrementing of the index into the buffer is a **critical section**.

P₁ (Σ)

P₂ (Σ)

Sem-wait();

Sem-wait();

put(i);

buffer[fill]=value;

race condition!

put(i);

buffer[fill]=value;

fill = fill + 1;

fill = fill + 1;

A Solution: Adding Mutual Exclusion

```
1  sem_t empty;
2  sem_t full;
3  sem_t mutex;
4
5  void *producer(void *arg) {
6      int i;
7      for (i = 0; i < loops; i++) {
8          sem_wait(&mutex);           // line p0 (NEW LINE)
9          sem_wait(&empty);           // line p1
10         put(i);                     // line p2
11         sem_post(&full);             // line p3
12         sem_post(&mutex);           // line p4 (NEW LINE)
13     }
14 }
15
```

Adding Mutual Exclusion (Incorrectly)

A Solution: Adding Mutual Exclusion

```
16 void *consumer(void *arg) {
17     int i;
18     for (i = 0; i < loops; i++) {
19         sem_wait(&mutex);           // line c0 (NEW LINE)
20         sem_wait(&full);           // line c1
21         int tmp = get();           // line c2
22         sem_post(&empty);          // line c3
23         sem_post(&mutex);          // line c4 (NEW LINE)
24         printf("%d\n", tmp);
25     }
26 }
```

Adding Mutual Exclusion (Incorrectly)

A Solution: Adding Mutual Exclusion (Cont.)

- Imagine two thread: one producer and one consumer.
 - The consumer **acquire** the `mutex` (line c0).
 - The consumer **calls** `sem_wait()` on the `full` semaphore (line c1).
 - The consumer is **blocked** and **yield** the CPU.
 - The consumer still holds the mutex!
 - The producer **calls** `sem_wait()` on the binary `mutex` semaphore (line p0).
 - The producer is now **stuck** waiting too. a classic deadlock.

(P₁)

sem-wait (mutex);
sem-wait (empty);
Z^{ZZ}

(C₁)

sem-wait (mutex);
Z^{ZZ}

mutex	empty
1	0
0	0
0	0
-1	0

A Working Solution

```
1  sem_t empty;
2  sem_t full;
3  sem_t mutex;
4
5  void *producer(void *arg) {
6      int i;
7      for (i = 0; i < loops; i++) {
8          sem_wait(&empty);           // line p1
9          sem_wait(&mutex);           // line p1.5 (MOVED MUTEX HERE...)
10         put(i);                     // line p2
11         sem_post(&mutex);           // line p2.5 (... AND HERE)
12         sem_post(&full);            // line p3
13     }
14 }
15
```

Grab the lock only when the next step is to access the shared variable (only when absolutely necessary)

Adding Mutual Exclusion (Correctly)

A Working Solution

```
16  void *consumer(void *arg) {
17      int i;
18      for (i = 0; i < loops; i++) {
19          sem_wait(&full);           // line c1
20          sem_wait(&mutex);          // line c1.5 (MOVED MUTEX HERE...)
21          int tmp = get();           // line c2
22          sem_post(&mutex);          // line c2.5 (... AND HERE)
23          sem_post(&empty);          // line c3
24          printf("%d\n", tmp);
25      }
26  }
27
28  int main(int argc, char *argv[]) {
29      // ...
30      sem_init(&empty, 0, MAX); // MAX buffers are empty to begin with ...
31      sem_init(&full, 0, 0);    // ... and 0 are full
32      sem_init(&mutex, 0, 1);   // mutex=1 because it is a lock
33      // ...
34  }
35      ...
```

Adding Mutual Exclusion (Correctly)

Reader-Writer Locks

- ▣ Imagine a number of concurrent list operations, including **inserts** and simple **lookups**.
 - ◆ **insert:**
 - Change the state of the list
 - A traditional critical section makes sense.
 - ◆ **lookup:**
 - Simply *read* the data structure.
 - As long as we can guarantee that no insert is on-going, we can allow many lookups to proceed **concurrently**.

This special type of lock is known as a **reader-writer lock**.

Reader-Writer Locks

- ❑ Only a **single writer** can acquire the lock.
- ❑ Once a reader has acquired a **read lock**,
 - ◆ **More readers** will be allowed to acquire the read lock too.
 - ◆ A writer will have to wait until all readers are finished.

```
1  typedef struct _rwlock_t {
2      sem_t lock;           // binary semaphore (basic lock)
3      sem_t writelock;      // used to allow ONE writer or MANY readers
4      int readers;          // count of readers reading in critical section
5  } rwlock_t;
```

```
1. void rwlock_init(rwlock_t *rw) {
2.     rw->readers = 0;
3.     sem_init(&rw->lock, 0, 1);
4.     sem_init(&rw->writelock, 0, 1);
5. }
```


Reader-Writer Locks (Cont.)

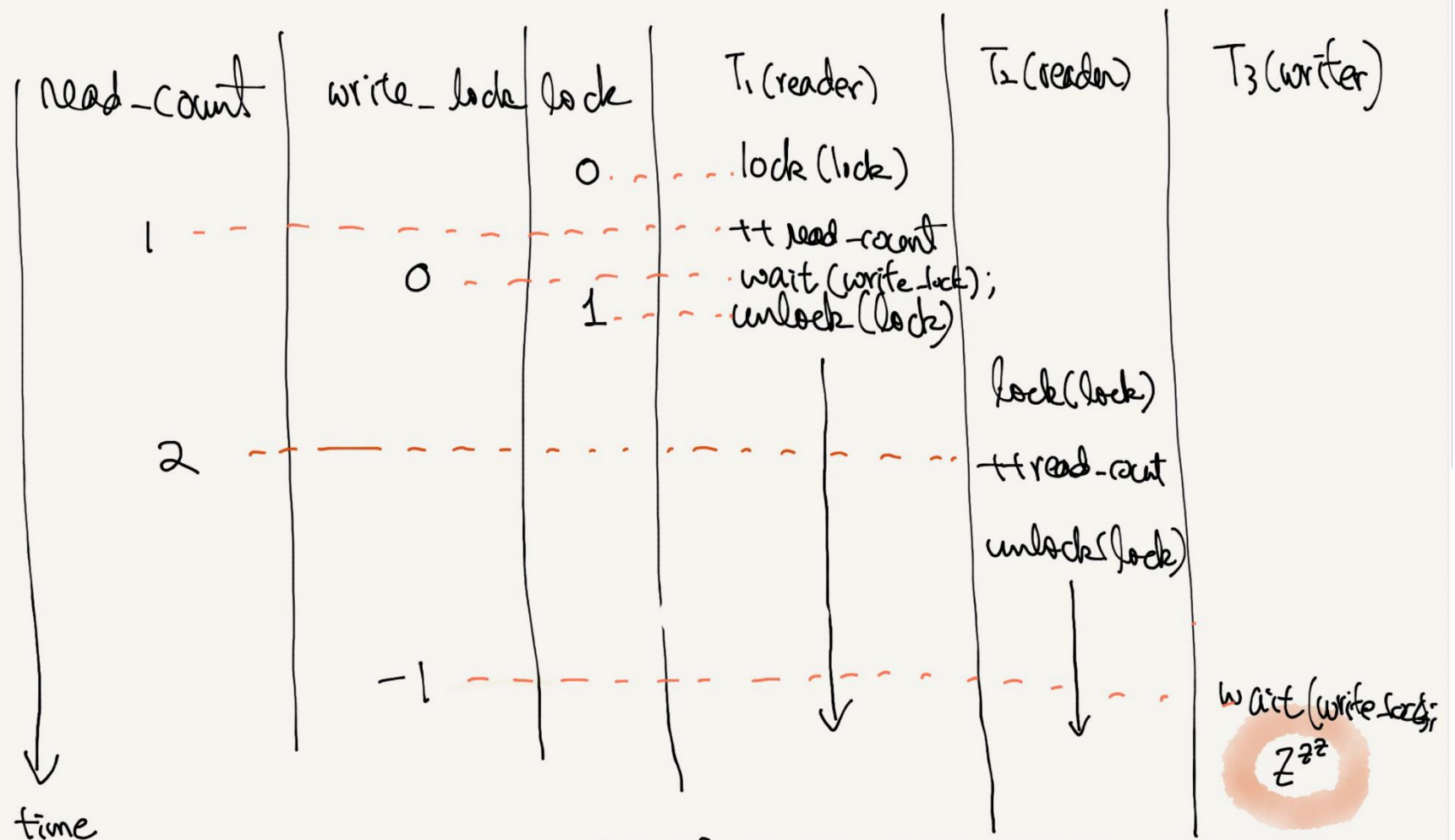
```
1. void rwlock_acquire_readlock(rwlock_t *rw) {
2.     sem_wait(&rw->lock); //critical section
3.     rw->readers++;
4.     if (rw->readers == 1)
5.         sem_wait(&rw->writelock); // first reader acquires writelock
6.     sem_post(&rw->lock);
7. }
```

```
1. void rwlock_release_readlock(rwlock_t *rw) {
2.     sem_wait(&rw->lock);
3.     rw->readers--;
4.     if (rw->readers == 0)
5.         sem_post(&rw->writelock); // last reader releases writelock
6.     sem_post(&rw->lock);
7. }
```

Reader-Writer Locks (Cont.)

```
1. void rwlock_acquire_writelock(rwlock_t *rw) {  
2.     sem_wait(&rw->writelock);  
3. }
```

```
1. void rwlock_release_writelock(rwlock_t *rw) {  
2.     sem_post(&rw->writelock);  
3. }
```



what if there comes new reader?

Reader-Writer Locks (Cont.)

- ▣ The reader-writer locks have **fairness problem**.
 - ◆ It would be relatively easy for reader to **starve writer**.
 - ◆ How to prevent more readers from entering the lock once a writer is waiting?

Reader-Writer Locks (Cont.)

- Add another semaphore to block new readers once a writer is waiting

```
#include <semaphore.h>
```

```
typedef struct {  
    sem_t mutex;           // Protects state (binary semaphore)  
    sem_t rw_mutex;        // Exclusive access for writers  
    sem_t queue;           // FIFO ordering gate  
  
    int active_readers;    // Currently reading  
    int waiting_writers;   // Writers waiting (protected by mutex)  
} sem_rwlock_t;
```

```
// Initialize
```

```
void sem_rwlock_init(sem_rwlock_t *rw) {  
    sem_init(&rw->mutex, 0, 1);  
    sem_init(&rw->rw_mutex, 0, 1);  
    sem_init(&rw->queue, 0, 1);  
    rw->active_readers = 0;  
    rw->waiting_writers = 0;  
}
```

```
// Reader lock
void sem_read_lock(sem_rwlock_t *rw) {
    sem_wait(&rw->queue);      // FIFO gate: line up here
    sem_wait(&rw->mutex);      // Enter critical section

    rw->active_readers++;
    if (rw->active_readers == 1) {
        sem_wait(&rw->rw_mutex); // First reader locks out writers
    }

    sem_post(&rw->mutex);      // Exit critical section
    sem_post(&rw->queue);      // Release FIFO gate for next
}

// Reader unlock
void sem_read_unlock(sem_rwlock_t *rw) {
    sem_wait(&rw->mutex);

    rw->active_readers--;
    if (rw->active_readers == 0) {
        sem_post(&rw->rw_mutex); // Last reader allows writers
    }

    sem_post(&rw->mutex);
}
```

```

// Writer lock
void sem_write_lock(sem_rwlock_t *rw) {
    sem_wait(&rw->queue);      // FIFO gate: blocks behind waiting
    readers/writers
    sem_wait(&rw->mutex);

    rw->waiting_writers++;      // Mark writer intent

    sem_post(&rw->mutex);

    // Must hold queue until we get rw_mutex to block new readers
    sem_wait(&rw->rw_mutex);    // Wait for all readers to exit

    sem_wait(&rw->mutex);
    rw->waiting_writers--;      // No longer waiting, now active
    sem_post(&rw->mutex);

    sem_post(&rw->queue);      // Release FIFO gate
}

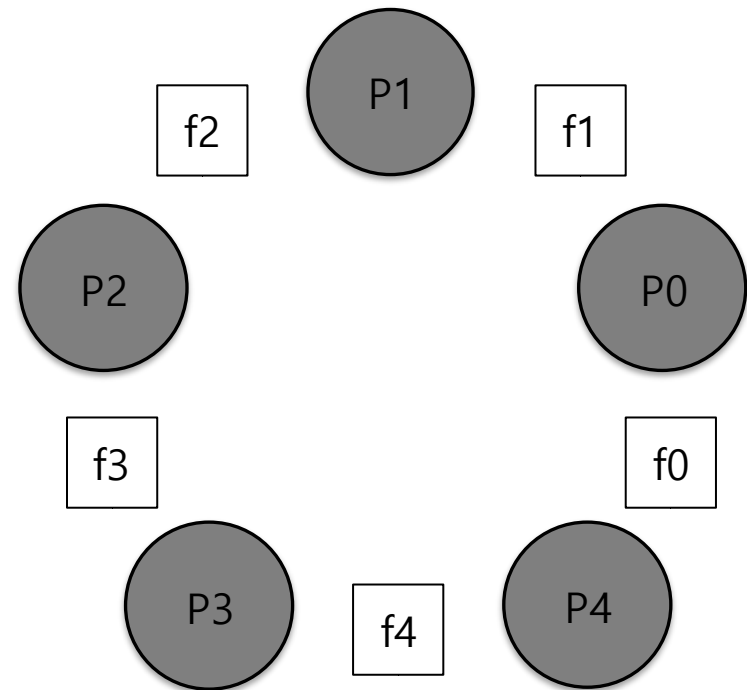
// Writer unlock
void sem_write_unlock(sem_rwlock_t *rw) {
    sem_post(&rw->rw_mutex);
}

```

**But, readers will be blocked
by queue as well!**

The Dining Philosophers

- ▣ Assume there are five “**philosophers**” sitting around a table.
 - ◆ Between each pair of philosophers is a single fork (five total).
 - ◆ The philosophers each have times when they **think**, and don’t need any forks, and times when they **eat**.
 - ◆ In order to eat, a philosopher needs **two forks**, both the one on their *left* and the one on their *right* (I know it’s weird...)
 - ◆ **Contention for these forks arises!**



The Dining Philosophers (Cont.)

▣ Key challenge

- ◆ There is **no deadlock**.
- ◆ **No** philosopher **starves** and never gets to eat.
- ◆ **Concurrency** is high.

```
while (1) {  
    think();  
    getforks();  
    eat();  
    putforks();  
}
```

Basic loop of each philosopher

```
// helper functions  
int left(int p) { return p; }  
  
int right(int p) {  
    return (p + 1) % 5;  
}
```

Helper functions (Downey's solutions)

- Philosopher p wishes to refer to the fork on their left $\rightarrow$ call `left(p)`.
- Philosopher p wishes to refer to the fork on their right $\rightarrow$ call `right(p)`.

The Dining Philosophers (Cont.)

- We need some **semaphores**, one for each fork: `sem_t forks[5]`.

```
1  void getforks() {
2      sem_wait(forks[left(p)]);
3      sem_wait(forks[right(p)]);
4  }
5
6  void putforks() {
7      sem_post(forks[left(p)]);
8      sem_post(forks[right(p)]);
9  }
```

The `getforks()` and `putforks()` Routines (Broken Solution)

- ◆ **Deadlock** occurs!
 - If each philosopher happens to **grab the fork on their left** before any philosopher can grab the fork on their right.
 - Each will be stuck *holding one fork* and waiting for another, *forever*.

A Solution: Breaking The Dependency

- Change how forks are acquired

- Let's assume that philosopher 4 acquire the forks in a *different order*.

```
1  void getforks() {  
2      if (p == 4) {  
3          sem_wait(forks[right(p)]);  
4          sem_wait(forks[left(p)]);  
5      } else {  
6          sem_wait(forks[left(p)]);  
7          sem_wait(forks[right(p)]);  
8      }  
9  }
```

- There is no situation where each philosopher grabs one fork and is stuck waiting for another. **The cycle of waiting is broken.**

Reference materials

How to Implement Semaphores using CV: Zemaphores

```
1  typedef struct __Zem_t {
2      int value;
3      pthread_cond_t cond;
4      pthread_mutex_t lock;
5  } Zem_t;
6
7  // only one thread can call this
8  void Zem_init(Zem_t *s, int value) {
9      s->value = value;
10     Cond_init(&s->cond);
11     Mutex_init(&s->lock);
12 }
```

Semaphore using condition variable: Zemaphores

```
1 void Zem_wait(Zem_t *s) {  
2     Mutex_lock(&s->lock);  
3     while (s->value <= 0)  
4         Cond_wait(&s->cond, &s->lock);  
5     s->value--;  
6     Mutex_unlock(&s->lock);  
7 }
```

How To Implement Semaphores (Cont.)

```
22  void Zem_post(Zem_t *s) {  
23      Mutex_lock(&s->lock);  
24      s->value++;  
25      Cond_signal(&s->cond);  
26      Mutex_unlock(&s->lock);  
27  }
```

- ◆ Zemaphores don't maintain the invariant that *the value of* the semaphore, when negative, reflects the number of waiting threads.
 - The value never be lower than zero.
 - This behavior is **easier** to implement and **matches** the current Linux implementation.

Using semaphores to implement C.V.

- ▣ It's much more difficult!
 - ◆ Try it yourself